Weijie Wu (Alan)

PhD Student · RS Foundation Models · Building Small Things

Email: wuweijie25@mails.ucas.ac.cn | GitHub: github.com/go-bananas-wwj | ModelScope: modelscope.cn/profile/WeijieWu

Education

University of Chinese Academy of Sciences Sep 2025 – Present

PhD, Signal and Information Processing | Direct PhD, advised by Prof. Guoqing Li | Aerospace Information Research Institute | Research: remote sensing foundation models, agents, geo-embedding

Chongqing University (Project 985) Sep 2021 – Jun 2025

Bachelor's, Artificial Intelligence | National School of Excellence in Engineering, Mingyue Innovation Class | Rank 1/15, GPA 3.79/4.0

Minor, Finance | School of Economics and Business Administration

Chongqing Nankai Secondary School Sep 2018 – Jun 2021

First Prize in Informatics Olympiad (Provincial)

Experience

Tsinghua University — Research Intern Mar 2024 – Sep 2024

- Research on automated game asset generation based on Diffusion Models
- Built complete workflow with ComfyUI, used ControlNet and character skeleton models to solve character consistency in image-to-video generation

Chongqing Mingyue Lake Innovation Base / X-bot Park — Founder Apr 2022 – Jul 2024

- Founded Cooling Technology project, developed a wearable intelligent cooling suit
- Led full 0-to-1 product cycle: mechanical design, circuit design, embedded development, prototyping, user research
- Collaborated with Chongqing Police College, secured 500K CNY seed funding from Liangjiang New Area government, achieved small-batch production and profitability

Ningbo Chujing Co., Ltd. — Embedded R&D Intern Jun 2023 – Aug 2023

- Developed embedded programs for intelligent mouthwash detection device
- Built ultrasonic module detection system

Projects

InstructSAM Oct 2024 – Present

A training-free framework for instruction-oriented remote sensing image segmentation. Users can precisely segment targets in RS images via natural language instructions without retraining.

- Stack: PyTorch, Vision-Language Model, SAM | NeurIPS 2025

Disaster-Agent May 2025 – Present

A multi-agent coordination platform for rapid disaster response, integrating real-time remote sensing data streams to support rapid damage assessment and decision assistance.

- Stack: Multi-Agent System, Remote Sensing, LLM

EarthEmbeddingExplorer 2024 – Present

A web application for cross-modal retrieval of global satellite images, optimizing visualization effects and user experience of geo-embeddings.

- Stack: Vue.js, Geo-Embedding, Cross-Modal Retrieval | ICLR Workshop

NODA Data Retrieval Assistant 2024 – Present

An intelligent data retrieval assistant for the NODA platform built on Dify, currently live in production.

- Stack: Dify, LLM, RAG | Link: noda.ac.cn

Wearable Cooling Suit Startup Mar 2023 – Jan 2024

- Led EVT development of cooling host unit (mechanical + circuit)
- Secured 500K CNY seed funding, completed full 0-to-1 product cycle

Publications

InstructSAM: A Training-Free Framework for Instruction-Oriented Remote Sensing Object Recognition

Zheng

EarthEmbeddingExplorer: A Web Application for Cross-Modal Retrieval of Global Satellite Images

Wu, W

Patents

Scalable Remote Sensing Deep Learning Sample Library Construction Method Based on Target Area Planning (CN120612563A)

Liquid Cooling Suit Related Patents (Multiple, Applied / Granted)

Awards

- Excellent Teaching Assistant (C Programming, C++ Programming), Chongqing University, Nov 2022 & Nov 2023
- Mathematical Contest in Modeling (MCM) National Third Prize, Jan 2023
- XbotPark Innovation Camp Best Project Award, Jun 2022 – Jul 2023
- Merit Student, Triple-A Student, and other honors (10+), Chongqing University
- Innovation & Entrepreneurship Competition Awards (10+), Chongqing University

Skills

Languages: Python, JavaScript, C++, MATLAB

Deep Learning: PyTorch, TensorFlow, Scikit-learn

Agents / LLM: Dify, LangChain, n8n, OpenClaw, Hermes Agent, Agent RAG

Frontend / Engineering: Vue.js, React, HTML, Git, Docker, Vite, Qt

Hardware / Embedded: ARM Cortex, Altium Designer, LCEDA, ROS

Design / Mechanical: SOLIDWORKS, Fusion360, ANSYS

Languages: Chinese (Native), English (CET-6 494, fluent in reading technical literature)